

DTSA 5303

# Ethical Issues in Data Science

|                                                 |                                        |               |
|-------------------------------------------------|----------------------------------------|---------------|
| FORMAT                                          | CATEGORY                               | CREDITS       |
| Online (DTSA)                                   | Core                                   | 1             |
| SECTION                                         | BEST ROLE MATCH                        | AVG ALIGNMENT |
| Section 2 – Vital Skills for Data Scientists    | Machine Learning Engineer (43.3 / 100) | 36.6 / 100    |
| SPECIALIZATION Vital Skills for Data Scientists |                                        |               |

37

CURRENT AVG SCORE  
across 6 job roles

70

POTENTIAL SCORE  
with recommended additions

+44

MAX SCORE GAIN  
best single-role improvement

## CURRENT COURSE DESCRIPTION

This course examines ethical issues related to data science, with the objective of making data science professionals aware of and sensitive to ethical considerations that may arise in their careers. It focuses on ethical frameworks, data science applications that lead to ethical considerations, current media and scholarly articles, and the perspectives and experiences of fellow students and computing professionals.

## INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE             | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|-----------------------------|------------|----------|
| Machine Learning Engineer | <div><div></div></div> 43.3 | <div><div></div></div> 68.4 | +25.1      | MEDIUM   |
| AI Engineer               | <div><div></div></div> 38.9 | <div><div></div></div> 75.1 | +36.2      | HIGH     |
| Data Scientist            | <div><div></div></div> 36.8 | <div><div></div></div> 61.4 | +24.6      | MEDIUM   |
| ML Engineer               | <div><div></div></div> 36.4 | <div><div></div></div> 66.4 | +30.0      | HIGH     |
| MLOps Engineer            | <div><div></div></div> 35.6 | <div><div></div></div> 77.9 | +42.3      | HIGH     |
| Applied Scientist         | <div><div></div></div> 28.8 | <div><div></div></div> 72.8 | +43.9      | HIGH     |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**System Design**

5 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

**Add System Design coverage** — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

|                                                                            |                         |
|----------------------------------------------------------------------------|-------------------------|
| <b>Machine Learning Engineer</b>                                           | 43.3 → 68.4 (+25.1 pts) |
| learning tools design models model machine technical systems               |                         |
| <b>AI Engineer</b>                                                         | 38.9 → 75.1 (+36.2 pts) |
| robotics models learning technical model systems teams problems            |                         |
| <b>Data Scientist</b>                                                      | 36.8 → 61.4 (+24.6 pts) |
| social people problems                                                     |                         |
| <b>ML Engineer</b>                                                         | 36.4 → 66.4 (+30.0 pts) |
| learning machine systems models model world problems impact                |                         |
| <b>MLOps Engineer</b>                                                      | 35.6 → 77.9 (+42.3 pts) |
| detection model scientists systems learning machine image tools            |                         |
| <b>Applied Scientist</b>                                                   | 28.8 → 72.8 (+43.9 pts) |
| computational research problems models technical learning systems industry |                         |

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.